Research analysis: Body Mass Index

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1. Paper metadata

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2. Abstract and overview

• BMI is a poor indicator of the percent of body fat and fails to capture information on the mass of fat in different body sites. [from the paper; verified; p. 117]

• A critical brief review discussing issues related to BMI, body fatness, and health. [from the paper; partially supported; p. 117]

• Current evidence demonstrates a wide range of BMIs over which mortality risk is modest, an effect that is age related. [from the paper; verified; p. 117]

• The paper provides a review addressing the limitations of BMI, body site fat mass, and mortality risk across BMI ranges. [from the paper; partially supported; p. 117]

3. Research problem

• BMI is an inaccurate indicator of percent body fat, does not differentiate between lean mass and fat mass, and fails to capture body fat distribution. [from the paper; verified; p. 117, p. 120]

• Anthropometric classification in adults, population-based epidemiological studies, public health policy, and risk assessment for health issues. [from the paper; partially supported; p. 117, p. 120]

• BMI categorization influences public health policies, medical risk assessments for chronic diseases, and severe social consequences such as discrimination and stigmatization. [from the paper; partially supported; p. 117]

• Simple and accurate methods for measuring percent body fat and specific regional fat depots are not readily available, and BMI does not reflect actual fat accumulation or anatomical distribution. [from the paper; verified; p. 122, p. 120]

4. Objectives

• To provide a critical review of BMI, obesity, and health, examining the limitations of BMI as a metric for fatness and mortality risk. [from the paper; partially supported; p. 117]

• Discuss how BMI fails to capture information on the mass of fat in different body sites and its connection to health and social issues. [from the paper; verified; p. 117]

• Discuss current evidence regarding the wide range of BMIs over which mortality risk is modest and how this relates to age. [from the paper; verified; p. 117]

• Review who Dr Quetelet was and how the Quetelet Index was derived. [from the paper; verified; p. 118]

5. Motivation

• BMI is widely accepted and used to guide public policy and health risk despite increasingly clear evidence that it is a poor indicator of body fat percentage. [from the paper; partially supported; p. 117]

• A critical reassessment and potential upward adjustment of BMI categories to accommodate population-based changes in height and weight and improvements in mortality rates. [from the paper; verified; p. 121]

• BMI does not differentiate between lean body mass and fat mass. [from the paper; verified; p. 120]

• BMI fails to provide information about the anatomical location or distribution of body fat depots. [from the paper; verified; p. 121]

• Correlations of mortality with BMI often fail to take into account confounding variables like familial genetics, disease history, or fitness. [from the paper; partially supported; p. 122]

6. Background

• Historically, weight adjusted for height emerged from life insurance data to assess life expectancy. Quetelet's index (weight divided by height squared) was later popularized by Keys et al. as the body mass index (BMI) and adopted by the WHO for standardized population categorization. [from the paper; partially supported; p. 118, p. 119]

• body weight, adjusted for height (Wt/Ht) / Metropolitan Life weight/height tables: Tables of average body weights for heights by gender and age used as a reference to determine underweight, overweight, and desirable body build. [from the paper; partially supported; p. 118]

• Quetelet Index / Body Mass Index (BMI): Weight in kilograms divided by height in meters squared; normalizes body mass distribution by reducing the effect of height variance. [from the paper; partially supported; p. 118, p. 119]

• Garrow Classification: Categorization of BMI into desirable (up to 25), grade I obesity (25-29.9), grade II obesity (30-40), and grade III obesity (>40). [from the paper; verified; p. 119]

• World Health Organization (WHO) classification: Uniform classification dividing BMI into four categories: underweight (15-19.9), normal weight (20-24.9), overweight (25-29.9), and obese (30-35 or greater). [from the paper; partially supported; p. 119]

• International Obesity Task Force classification: An expansion of BMI categories using preobesity (25 to 29.9), class I obesity (30 to 34.9), class II obesity (34.9 to 39.9), and class III obesity (40 or greater). [from the paper; partially supported; p. 120]

7. Methodology

• A critical narrative and historical review evaluating the development, mathematical basis, epidemiological utility, and physiological limitations of BMI (Quetelet Index) and alternative methods for estimating body fat mass and depot locations. [from the paper; verified; p. 117, p. 124]

• Historical examination of early weight-for-height tables, mathematical analysis of scaling height and weight (from cube formulas to Wt/Ht^2), review of adoption and categorical classifications (WHO, Garrow, NIH), critique of BMI's correlation with true body fat, and evaluation of alternate direct and indirect body fat measurement technologies. [from the paper; partially supported; p. 118, p. 119, p. 122]

• Evaluation of historical weight-for-height tables and the associated frame size adjustments and limitations. [from the paper; partially supported; p. 118]

• Mathematical scaling analysis of height and body mass (derivation and adoption of the Quetelet Index / BMI). [from the paper; verified; p. 118, p. 119]

• Analysis of categorical classifications of BMI by organizations including WHO, NIH, and Garrow. [from the paper; partially supported; p. 119]

• Critical review of modern direct and indirect measurement methods for total body fat mass and fat depot locations. [from the paper; verified; p. 122]

• Calculation of BMI via dividing body weight in kilograms by height in meters squared (kg/m^2) to normalize body mass distribution across heights. [from the paper; partially supported; p. 119]

• The paper is a critical review synthesizing existing published literature, historical studies, and epidemiological survey data rather than presenting a novel primary experiment. [from the paper; partially supported; p. 117]

8. Algorithms and models

• The paper uses Body Mass Index / Quetelet Index (Primary metric analyzed). [from the paper; verified; p. 118, p. 119]

• The paper uses Exponential height-scaling model (Wt/Ht^1.6 to Wt/Ht^1.7) (Comparative scaling formula). [from the paper; verified; p. 118]

• Height exponent = 1.6 to 1.7 (Body mass for height scaling model) [from the paper; verified; p. 118]

• The paper uses Volumetric 3-dimensional models (cube root of weight / height; weight / height^3) (Historical comparison model). [from the paper; verified; p. 118]

• The paper uses WHO BMI Classification System (Standard international classification reference). [from the paper; unsupported; no located source]

• underweight cutoff range = 15 to 19.9 (WHO BMI Classification) [from the paper; verified; p. 119]

• normal weight cutoff range = 20 to 24.9 (WHO BMI Classification) [from the paper; verified; p. 119]

• overweight cutoff range = 25 to 29.9 (WHO BMI Classification) [from the paper; verified; p. 119]

• obese cutoff range = 30 to 35 or greater (WHO BMI Classification) [from the paper; verified; p. 119]

• The paper uses Garrow Classification System (Historical categorization model). [from the paper; verified; p. 119]

• desirable cutoff = up to 25 (Garrow Classification) [from the paper; partially supported; p. 119]

• grade I obesity cutoff range = 25 to 29.9 (Garrow Classification) [from the paper; verified; p. 119]

• grade II obesity cutoff range = 30 to 40 (Garrow Classification) [from the paper; verified; p. 119]

• grade III obesity cutoff = greater than 40 (Garrow Classification) [from the paper; verified; p. 119]

9. Dataset and experimental setup

• Dataset: EPIC observational study; source Europe; size 359 387 individuals [from the paper; verified; p. 123]

• Dataset: Rosenbaum and colleagues survey; source Great Britain [from the paper; verified; p. 119]

• Dataset: National Health and Nutrition Examination Study (NHANES) II [from the paper; verified; p. 120]

• Critical review synthesizing historical data, physiological studies, population surveys, and observational cohorts. [from the paper; partially supported; p. 117]

10. Results and findings

• Percentage = 67.6 % (Desirable (BMI up to 25), Women; Rosenbaum and colleagues survey in Great Britain (1985)): Distribution percentage of women in the Desirable category (BMI up to 25) under the Garrow Classification. [from the paper; partially supported; p. 119]

• Percentage = 58.2 % (Desirable (BMI up to 25), Men; Rosenbaum and colleagues survey in Great Britain (1985)): Distribution percentage of men in the Desirable category (BMI up to 25) under the Garrow Classification. [from the paper; partially supported; p. 119]

• BMI = 16.9 (Men, complete absence of body fat stores): BMI value calculated by linear regression representing a complete absence of body fat stores in men. [from the paper; verified; p. 119]

• BMI = 13.7 (Women, complete absence of body fat stores): BMI value calculated by linear regression representing a complete absence of body fat stores in women. [from the paper; verified; p. 119]

• Both crude and adjusted relative risk of death among men was lowest in those with a BMI of 26.5 to 28. [from the paper; verified; p. 123]

• Risk of death factor increase = 1.17 factor (Men, 5-cm increase in circumference; EPIC observational study; baseline: Given BMI baseline): Factor increase in risk of death for any given BMI per 5-cm increase in circumference among men. [from the paper; partially supported; p. 123]

• Risk of death factor increase = 1.13 factor (Women, 5-cm increase in circumference; EPIC observational study; baseline: Given BMI baseline): Factor increase in risk of death for any given BMI per 5-cm increase in circumference among women. [from the paper; partially supported; p. 123]

• BMI has serious limitations and can be misleading when used as an indicator of percent of body fat mass, especially in men. [author's claim; verified; p. 124]

11. Research contributions

• A critical review of BMI, obesity, and health, discussing why BMI is a poor indicator of body fat percentage and fat distribution, and showing that mortality risk is modest over a wide range of BMIs. [author's claim; verified; p. 117]

• Synthesizes historical and clinical evidence to caution against the exclusive clinical and public health reliance on BMI cutoffs for health risk evaluation and intervention design. [SYSTEM INFERENCE; partially supported; p. 124]

12. Limitations

• BMI is a rather poor indicator of percent of body fat and does not capture information on the mass of fat in different body sites. [author's claim; verified; p. 117]

• Simple and accurate methods for measuring percent of body fat and specific body fat depots are not available. [author's claim; verified; p. 122]

• Population-based studies correlating mortality rates with BMI frequently fail to consider family history, smoking, alcohol abuse, duration or type of obesity, occupation, treatment status, and changes in weight/height with age. [author's claim; partially supported; p. 122]

13. Future scope

• Moving beyond BMI as a surrogate for body fat mass, or updating its categories and definitions to reflect the current general population distribution. [author's claim; verified; p. 124]

• Developing a better means than BMI for estimating percent of body fat and its relationship to mortality and morbidities. [author's claim; verified; p. 124]

• Resolving whether BMI, total body fat mass, or fat distribution (visceral vs subcutaneous, abdominal circumference, waist/hip ratio) is more important in predicting comorbidities and mortality. [author's claim; verified; p. 124]

14. Technical methodology breakdown

• Step 1: Calculate Body Mass Index (BMI) [from the paper; partially supported; p. 119]

• Step 2: Assign Weight/Obesity Category [from the paper; partially supported; p. 119]

• Step 3: Direct/Indirect Assessment of Body Fat and Fat Depots [from the paper; partially supported; p. 122]

15. Technical terminology

• Body Mass Index (BMI): The metric currently in use for defining anthropometric height/weight characteristics in adults and for classifying (categorizing) them into groups; calculated as body weight (kilograms) divided by height squared (meters). [from the paper; verified; p. 117, p. 119]

• Quetelet Index: Body weight (kilograms) divided by height squared (meters). [from the paper; verified; p. 118, p. 119]

• Obesity: An excessive accumulation of body fat (triacylglycerols). [from the paper; verified; p. 120]

• Desirable weight: The ratios of weights for heights (or weight/height) representing the lowest mortality in adults. [from the paper; verified; p. 118]

• Dual-energy x-ray absorptiometry (DEXA): A scan that provides a 3-dimensional picture of body organ densities, used for estimating total body fat and its location. [from the paper; partially supported; p. 122]

• Computed tomography (CT): An imaging technique where slices of body parts (such as abdomen and thigh) are used to obtain 2 dimensions of fat depots to reconstruct a 3-dimensional fat area. [SYSTEM EXPLANATION; verified; p. 122]

16. Verification summary

not verifiable: 1

partially supported: 29

unsupported: 1

verified: 49

17. Evidence and page references

EVD_001 (CLM_001), p. 117, section unknown: “Body Mass Index”

EVD_002 (CLM_002), p. 117, section unknown: “DOI: 10.1097/NT.0000000000000092”

EVD_003 (CLM_004), p. 117, Front matter: “However, it is increasingly clear that BMI is a rather poor indicator of percent of body fat. Importantly, the BMI also does not capture information on the mass of fat in different body sites.”

EVD_004 (CLM_005), p. 117, Front matter: “All of these issues are discussed in this brief review.”

EVD_005 (CLM_006), p. 117, Front matter: “current evidence indicates there is a wide range of BMIs over which mortality risk is modest, and this is age related.”

EVD_006 (CLM_007), p. 117, Front matter: “All of these issues are discussed in this brief review.”

EVD_007 (CLM_008), p. 117, Front matter: “However, it is increasingly clear that BMI is a rather poor indicator of percent of body fat. Importantly, the BMI also does not capture information on the mass of fat in different body sites.”

EVD_008 (CLM_008), p. 120, Front matter: “A particular problem with BMI as an index of obesity is that it does not differentiate between body lean mass and body fat mass”

EVD_009 (CLM_009), p. 117, Front matter: “The body mass index (BMI) is the metric currently in use for defining anthropometric height/weight characteristics in adults and for classifying (categorizing) them into groups.”

EVD_010 (CLM_009), p. 120, Front matter: “at present the BMI is the currency by which we define the obesity issue throughout the Western world. It was developed for the convenience of the epidemiologists”

EVD_011 (CLM_010), p. 117, Front matter: “It also is widely used as a risk factor for the development of or the prevalence of several health issues. In addition, it is widely used in determining public health policies.”

EVD_012 (CLM_010), p. 117, Front matter: “The social consequences of being ‘‘too fat’’ are severe. Discrimination begins in childhood and results in serious emotional scars.”

EVD_013 (CLM_011), p. 122, Front matter: “At present, simple, accurate methods for measuring percent of body fat and, in particular, body fat in different fat depots are not available.”

EVD_014 (CLM_011), p. 120, Front matter: “From an anatomical and metabolic perspective, the term obesity should refer to an excessive accumulation of body fat (triacylglycerols), and upon these grounds, the accuracy of the BMI as a determinant of body fat mass has been repeatedly questioned”

EVD_015 (CLM_012), p. 117, Front matter: “Obesity, BMI, and Health: A Critical Review”

EVD_016 (CLM_012), p. 117, Front matter: “All of these issues are discussed in this brief review.”

EVD_017 (CLM_013), p. 117, Front matter: “All of these issues are discussed in this brief review.”

EVD_018 (CLM_013), p. 117, Front matter: “Importantly, the BMI also does not capture information on the mass of fat in different body sites. The latter is related not only to untoward health issues but to social issues as well.”

EVD_019 (CLM_014), p. 117, Front matter: “Lastly, current evidence indicates there is a wide range of BMIs over which mortality risk is modest, and this is age related. All of these issues are discussed in this brief review.”

EVD_020 (CLM_015), p. 118, Front matter: “I would like to briefly mention who Dr Quetelet was and how the ‘‘Quetelet Index’’ was derived.”

EVD_021 (CLM_016), p. 117, Front matter: “The body mass index (BMI) is the metric currently in use for defining anthropometric height/weight characteristics in adults and for classifying (categorizing) them into groups. The common interpretation is that it represents an index of an individual’s fatness.”

EVD_022 (CLM_016), p. 117, Front matter: “However, it is increasingly clear that BMI is a rather poor indicator of percent of body fat.”

EVD_023 (CLM_017), p. 121, Front matter: “These data suggest that the BMI categories should be adjusted upward periodically to accommodate population-based changes. Improvements in mortality rates also suggest an adjustment would be useful.”

EVD_024 (CLM_018), p. 120, Front matter: “A particular problem with BMI as an index of obesity is that it does not differentiate between body lean mass and body fat mass; that is, a person can have a high BMI but still have a very low fat mass and vice versa.”

EVD_025 (CLM_019), p. 121, Front matter: “An additional limitation of the BMI is it does not capture body fat location information. This is an important variable in assessing the metabolic as well as mortality consequences of excessive fat accumulation.”

EVD_026 (CLM_020), p. 122, Front matter: “Even when some comorbidities are considered, the correlation of mortality rates with BMI often does not take into consideration such factors as family history of diabetes, hypertension, coronary heart disease, metabolic syndrome, dyslipidemias,

familial longevity or the family prevalence of carcinomas, and so on.”

EVD_027 (CLM_021), p. 118, Front matter: “Life insurance data accumulated at that time5 and subsequently6 indicated that body weight, adjusted for height (Wt/Ht), was an independent determinant of life expectancy”

EVD_028 (CLM_021), p. 119, Front matter: “Keys et al, using better documented weight for height data, popularized the Quetelet Index in population-based studies. They referred to it as the body mass index (BMI).”

EVD_029 (CLM_022), p. 118, Front matter: “Life insurance data accumulated at that time5 and subsequently6 indicated that body weight, adjusted for height (Wt/Ht), was an independent determinant of life expectancy”

EVD_030 (CLM_022), p. 118, Front matter: “Subsequently, the Metropolitan Life Insurance Company in 1959 published tables of average body weights for heights (Wt/Ht) by gender and at different ages.7”

EVD_031 (CLM_023), p. 118, Front matter: “This exponent is not convenient for use in populationbased studies, and it was determined that Wt/Ht2 generally was satisfactory.16,18 The latter represents the Quetelet Index.”

EVD_032 (CLM_023), p. 119, Front matter: “Instead, Keys et al, using better documented weight for height data, popularized the Quetelet Index in population-based studies. They referred to it as the body mass index (BMI).”

EVD_033 (CLM_024), p. 119, Front matter: “Theabove4categoriesaresimilartothosesuggestedbyJohnS. Garrow in 1981,31,32 but the terminology was changed. The terminologyheusedwas‘‘desirable’’foraBMIupto25,‘‘grade I obesity’’ between 25 and 29.9, ‘‘grade II obesity,’’ between 30 and 40, and ‘‘grade III obesity’’ for BMI greater than 40.”

EVD_034 (CLM_025), p. 119, Front matter: “In 1993, the WHO assembled an Expert Consultation Group with a charge of developing uniform categories of the BMI. The results were published as a technical report in 1995.25”

EVD_035 (CLM_025), p. 119, Front matter: “Four categories were established: underweight, normal, overweight, and obese.”

EVD_036 (CLM_026), p. 120, Front matter: “In 1997, the International Obesity Task Force expanded the number of BMI categories to include different degrees of obesity and changed the terminology modestly.36”

EVD_037 (CLM_027), p. 117, Front matter: “Frank Q. Nuttall, MD, PhD”

EVD_038 (CLM_028), p. 117, Front matter: “Nutr Today”

EVD_039 (CLM_029), p. 117, Front matter: “2015;50(3):117Y128”

EVD_040 (CLM_030), p. 117, Front matter: “Obesity, BMI, and Health: A Critical Review Frank Q. Nuttall, MD, PhD The body mass index (BMI) is the metric currently in use for defining anthropometric height/weight characteristics in adults and for classifying (categorizing) them into groups.”

EVD_041 (CLM_030), p. 124, Front matter: “It should be understood that the BMI has serious limitations when used as an indicator of percent of body fat mass. Indeed, it may be misleading in this regard, particularly in men.”

EVD_042 (CLM_031), p. 118, Front matter: “Subsequently, the Metropolitan Life Insurance Company in 1959 published tables of average body weights for heights (Wt/Ht) by gender and at different ages.”

EVD_043 (CLM_031), p. 118, Front matter: “Mathematically, the issue of adjusting body build for differences in height was approached with the concept that the body, particularly the trunk, could be considered as being a 3-dimensional volume or mass.”

EVD_044 (CLM_031), p. 119, Front matter: “Instead, Keys et al, using better documented weight for height data, popularized the Quetelet Index in population-based studies. They referred to it as the body mass index (BMI).”

EVD_045 (CLM_031), p. 122, Front matter: “Methods of Estimating Body Fat Mass and Location of the Fat At present, simple, accurate methods for measuring percent of body fat and, in particular, body fat in different fat depots are not available.”

EVD_046 (CLM_032), p. 118, Front matter: “Subsequently, the Metropolitan Life Insurance Company in 1959 published tables of average body weights for heights (Wt/Ht) by gender and at different ages.7”

EVD_047 (CLM_032), p. 118, Front matter: “Problems With the Wt/Ht (Body Build) Index Early on it was recognized that tall people had a lower death rate than did short people7,8,13 with the same Wt/Ht ratio.”

EVD_048 (CLM_033), p. 118, Front matter: “Mathematically, the issue of adjusting body build for differences in height was approached with the concept that the body, particularly the trunk, could be considered as being a 3-dimensional volume or mass.”

EVD_049 (CLM_033), p. 118, Front matter: “This exponent is not convenient for use in populationbased studies, and it was determined that Wt/Ht2 generally was satisfactory.16,18 The latter represents the Quetelet Index.”

EVD_050 (CLM_033), p. 119, Front matter: “Keys et al, using better documented weight for height data, popularized the Quetelet Index in population-based studies. They referred to it as the body mass index (BMI).”

EVD_051 (CLM_034), p. 119, Front matter: “In 1993, the WHO assembled an Expert Consultation Group with a charge of developing uniform categories of the BMI. The results were published as a technical report in 1995.25”

EVD_052 (CLM_034), p. 119, Front matter: “Four categories were established: underweight, normal, overweight, and obese.”

EVD_053 (CLM_035), p. 122, Front matter: “Methods of Estimating Body Fat Mass and Location of the Fat At present, simple, accurate methods for measuring percent of body fat and, in particular, body fat in different fat depots are not available.”

EVD_054 (CLM_035), p. 122, Front matter: “All of the previously mentioned methods use certain assumptions in the calculation of body fat mass, and all are subject to potential error. Nevertheless, there are more specific methods of determining body fat mass than is the BMI.”

EVD_055 (CLM_036), p. 119, Front matter: “Thus, Quetelet Index = body weight (kilograms) divided by height squared (meters) = BMI. As indicated above, by squaring the height, it reduces the contribution of leg length in the equation and tends to normalize the bodymass distribution at each level of height”

EVD_056 (CLM_037), p. 117, Front matter: “Obesity, BMI, and Health: A Critical Review Frank Q. Nuttall, MD, PhD”

EVD_057 (CLM_038), p. 118, Front matter: “it was determined that Wt/Ht2 generally was satisfactory.16,18 The latter represents the Quetelet Index.”

EVD_058 (CLM_038), p. 119, Front matter: “Thus, Quetelet Index = body weight (kilograms) divided by height squared (meters) = BMI.”

EVD_059 (CLM_039), p. 118, Front matter: “Later, it was shown that the body mass for height actually scaled best with weight for height when the height was raised to the 1.6 to 1.7 exponent (Wt/Ht,1.6 etc).18”

EVD_060 (CLM_040), p. 118, Front matter: “when the height was raised to the 1.6 to 1.7 exponent (Wt/Ht,1.6 etc).18”

EVD_061 (CLM_041), p. 118, Front matter: “Indeed, several equations were developed and tested based on this concept; that is, the cube root of weight divided by height (3¾Wt/Ht) or weight/height,3 and so on, but none were ideal.17”

EVD_063 (CLM_043), p. 119, Front matter: “underweight if his/her BMI was in the range of 15 to 19.9”

EVD_064 (CLM_044), p. 119, Front matter: “normal weight if the BMI was 20 to 24.9”

EVD_065 (CLM_045), p. 119, Front matter: “overweight if the BMI was 25 to 29.9”

EVD_066 (CLM_046), p. 119, Front matter: “obese if it was 30 to 35 or greater.”

EVD_067 (CLM_047), p. 119, Front matter: “The terminologyheusedwas‘‘desirable’’foraBMIupto25,‘‘grade I obesity’’ between 25 and 29.9, ‘‘grade II obesity,’’ between 30 and 40, and ‘‘grade III obesity’’ for BMI greater than 40.”

EVD_068 (CLM_048), p. 119, Front matter: “‘‘desirable’’foraBMIupto25”

EVD_069 (CLM_049), p. 119, Front matter: “‘‘grade I obesity’’ between 25 and 29.9”

EVD_070 (CLM_050), p. 119, Front matter: “‘‘grade II obesity,’’ between 30 and 40”

EVD_071 (CLM_051), p. 119, Front matter: “‘‘grade III obesity’’ for BMI greater than 40.”

EVD_072 (CLM_052), p. 119, Front matter: “Thus, Quetelet Index = body weight (kilograms) divided by height squared (meters) = BMI.”

EVD_073 (CLM_053), p. 119, Front matter: “Four categories were established: underweight, normal, overweight, and obese. An individual would be considered to be underweight if his/her BMI was in the range of 15 to 19.9, normal weight if the BMI was 20 to 24.9, overweight if the BMI was 25 to 29.9, and obese if it was 30 to 35 or greater.”

EVD_074 (CLM_054), p. 122, Front matter: “The indirect methods currently in use for estimating total percent of body fat include underwater weighing, an air displacement and density determination using a Bod Pod, a bioelectrical impedance analyzer, and a determination of the isotopically labeled water mass.”

EVD_075 (CLM_054), p. 122, Front matter: “Becauseoftheirconvenience,bioelectricimpedancemethods or DEXA scans are the most commonly used to estimate the amount and, with DEXA scans, the location of body fat depots.”

EVD_076 (CLM_055), p. 117, Front matter: “The body mass index (BMI) is the metric currently in use for defining anthropometric height/weight characteristics in adults and for classifying (categorizing) them into groups.”

EVD_077 (CLM_055), p. 119, Front matter: “Thus, Quetelet Index = body weight (kilograms) divided by height squared (meters) = BMI.”

EVD_078 (CLM_056), p. 118, Front matter: “it was determined that Wt/Ht2 generally was satisfactory.16,18 The latter represents the Quetelet Index.”

EVD_079 (CLM_056), p. 119, Front matter: “Thus, Quetelet Index = body weight (kilograms) divided by height squared (meters) = BMI.”

EVD_080 (CLM_057), p. 120, Front matter: “From an anatomical and metabolic perspective, the term obesity should refer to an excessive accumulation of body fat (triacylglycerols)”

EVD_081 (CLM_058), p. 118, Front matter: “The insurance data also indicated the ratios of weights for heights (the term used was ‘‘body build’’) at which mortality was lowest in adults. The latter was referred to as the ‘‘ideal’’ or later the ‘‘desirable’’ weight.”

EVD_082 (CLM_059), p. 122, Front matter: “A dual-energy x-ray absorptiometry (DEXA) scan, which provides a 3-dimensional picture of body organ densities, can be used for estimating total body fat.”

EVD_083 (CLM_060), p. 122, Front matter: “Single computed tomography (CT) slices of the abdomen and thigh can be used to obtain 2”

EVD_084 (CLM_060), p. 122, Front matter: “dimensions of those fat depots from which a 3-dimensional fat area can be reconstructed.”

EVD_085 (CLM_061), p. 123, Front matter: “The EPIC observational study is a population-based study that includes 359 387 individuals aged 25 to 70 years living in Europe.109 The mean age of this group at the initiation of the study was 51.5 years, and the mean follow-up has been 9.7 T 2 years.”

EVD_086 (CLM_062), p. 119, Front matter: “The latter classification was based on Rosenbaum and colleagues’33 own data obtained in a survey of an adult population,aged16to64years,inGreatBritainandpublishedin1985.”

EVD_087 (CLM_063), p. 120, Front matter: “This was based on an 85% cutoff point of people examined in the National Health and Nutrition Examination Study (NHANES) II.12,22,34”

EVD_088 (CLM_064), p. 117, Front matter: “All of these issues are discussed in this brief review.”

EVD_089 (CLM_065), p. 119, Front matter: “TABLE 1 Garrow Classification Obesity Desirable Grade I Grade II Grade III BMI (Up to 25) (25Y29.9) (30Y40) (Q40) Percentage Women 67.6 24 8 0.4 Men 58.2 34 6 0.2”

EVD_090 (CLM_066), p. 119, Front matter: “TABLE 1 Garrow Classification Obesity Desirable Grade I Grade II Grade III BMI (Up to 25) (25Y29.9) (30Y40) (Q40) Percentage Women 67.6 24 8 0.4 Men 58.2 34 6 0.2”

EVD_091 (CLM_067), p. 119, Front matter: “Using linear regression, a BMI of 16.9 in men and 13.7 in women represents a complete absence of body fat stores.31”

EVD_092 (CLM_068), p. 119, Front matter: “Using linear regression, a BMI of 16.9 in men and 13.7 in women represents a complete absence of body fat stores.31”

EVD_093 (CLM_069), p. 123, Front matter: “In this study, both the crude and ad- justed relative risk of death among men was actually the lowest in those with a BMI of 26.5 to 28, that is, those in the overweight (preobese) category.”

EVD_094 (CLM_070), p. 123, Front matter: “Indeed, for any given BMI, a 5-cm increase in circumference increased the risk of death by a factor of 1.17 among men and 1.13 for women.”

EVD_095 (CLM_071), p. 123, Front matter: “Indeed, for any given BMI, a 5-cm increase in circumference increased the risk of death by a factor of 1.17 among men and 1.13 for women.”

EVD_096 (CLM_072), p. 124, Front matter: “It should be understood that the BMI has serious limitations when used as an indicator of percent of body fat mass. Indeed, it may be misleading in this regard, particularly in men.”

EVD_097 (CLM_073), p. 117, Front matter: “However, it is increasingly clear that BMI is a rather poor indicator of percent of body fat. Importantly, the BMI also does not capture information on the mass of fat in different body sites.”

EVD_098 (CLM_074), p. 122, Front matter: “At present, simple, accurate methods for measuring percent of body fat and, in particular, body fat in different fat depots are not available.”

EVD_099 (CLM_075), p. 122, Front matter: “Even when some comorbidities are considered, the correlation of mortality rates with BMI often does not take into consideration such factors as family history of diabetes, hypertension, coronary heart disease, metabolic syndrome, dyslipidemias, familial longevity or the family prevalence of carcinomas, and so on.”

EVD_100 (CLM_075), p. 122, Front matter: “In most population-based studies, only the initial weight and/or BMI are given, even though weight as well as fat stores are known to increase and height to decrease with aging.”

EVD_101 (CLM_076), p. 124, Front matter: “It is time to move beyond the BMI as a surrogate for determining body fat mass. Alternatively, if BMI continues to be used, the categories and definitions

should be changed to reflect the current distribution of BMIs in the general population.”

EVD_102 (CLM_077), p. 124, Front matter: “A better means than the BMI for estimating percent of body fat and its relationship to mortality and various morbidities clearly would be desirable.”

EVD_103 (CLM_078), p. 124, Front matter: “Overall, a major unresolved issue is which factor of the following is more important in the prediction of comorbidities such as cardiovascular disease, diabetes, hypertension, malignancies, or overall death rates. Is it BMI, total body fat mass, or the distribution of body fat, that is, visceral versus subcutaneous, or upper body fat accumulation (as determined by abdominal circumference, or a waist/hip ratio, or some combination of these, and so on)?”

EVD_104 (CLM_079), p. 117, Front matter: “The BMI has been useful in populationbased studies by virtue of its wide acceptance in defining specific categories of body mass as a health issue. However, it is increasingly clear that BMI is a rather poor indicator of percent of body fat.”

EVD_105 (CLM_079), p. 117, Front matter: “Lastly, current evidence indicates there is a wide range of BMIs over which mortality risk is modest, and this is age related. All of these issues are discussed in this brief review.”

EVD_106 (CLM_080), p. 124, Front matter: “The BMI was not originally developed for use specifically as an index of fatness in population-based studies. However, it has been coopted for this use because it is a readily obtained metric.”

EVD_107 (CLM_080), p. 124, Front matter: “A BMI-determined categorization of an individual should not be used exclusively in counseling or in the design of a treatment regimen.”

Provenance

Model gemini-3.8-flash; prompt extraction-v3; schema 1.0; embedding sentence-transformers/multi-qa-mpnet-base-cos-v1; index 1.